

02.1 — Shift Management Workflow

Audience: HR managers, shift supervisors, hospital admin, anyone running day-to-day shift ops. **Goal:** End-to-end explanation of how Haritha Hospitals manages shifts — from entity setup to daily operations to payroll. **Last updated:** 2026-08-29 **Pre-req reading:** 00.1 What is ERP/Frappe/HRMS, 01.1 Schema Reference, 01.2 Schema Diagram

Overview

Shift Management at Haritha Hospitals is the system that defines when each of our 210+ employees works, records whether they actually showed up, and feeds that signal into payroll.

Hospitals don't run 9-to-5. Emergency, ICU, and ward staff work in rotating shifts that change week to week. Without a structured system you end up with: - Phone calls the night before to confirm "are you on duty?" - A register book at the nurses' station that gets lost - Payroll built from a manager's best guess - No way to audit who was on shift when an incident happened

Haritha uses **HRMS Shift Management** (an app that ships with Frappe) to replace all of this with: - **Shift Types** — defined shift timings (e.g., Morning 06:00–14:30, Night 20:00–08:00) - **Shift Assignments** — which employee works which shift on which day - **Employee Check-ins** — biometric / manual log of actual arrival and departure - **Attendance** — derived status (Present / Absent / On Leave / Half Day / Holiday) - **Salary Slip** — calculated from attendance + leave + salary structure

The end result: the manager looks at a **monthly Roster** grid, the HR team reviews **Attendance** records, and payroll reads the same attendance data to compute the month's pay — all from one database.

Entities recap

The schema is documented in detail in 01.1. Quick recap for this doc:

Entity	DocType	Role
Employee	HRMS Employee	Master record — name, dept, designation, default shift, bank, IDs
Department	HRMS Department	Org unit — owns employees, links to Cost Center
Designation	HRMS Designation	Job title (Nurse, Doctor, Housekeeping, Security)
Shift Type	HRMS Shift Type	Shift timing template (start/end, color, grace periods)
Shift Assignment	HRMS Shift Assignment	One row = "Employee X works Shift Type Y from date A to date B"
Attendance	HRMS Attendance	One row per employee per day — Present / Absent / On Leave / Half Day

Entity	DocType	Role
Employee Checkin	HRMS Employee Checkin	Raw IN/OUT log entries (from biometric or manual)
Holiday List	HRMS Holiday List	Calendar of org holidays (per Company)
Leave Application	HRMS Leave Application	Employee's request for paid/unpaid leave
Leave Type	HRMS Leave Type	Defines a kind of leave (Casual, Sick, Earned, etc.)
Leave Allocation	HRMS Leave Allocation	Per-employee annual quota per Leave Type

The Haritha custom app adds fields to several of these — see 01.1 §Custom Fields for the full list. The most shift-relevant are:

- Employee.default_shift (Link → Shift Type) — fallback for Auto Attendance cron
- Employee.shift_request_approver (Link → User)
- Employee.leave_approver (Link → User)
- Employee.expense_approver (Link → User)
- Company.default_payroll_payable_account (Link → Account)
- Department.payroll_cost_center (Link → Cost Center)

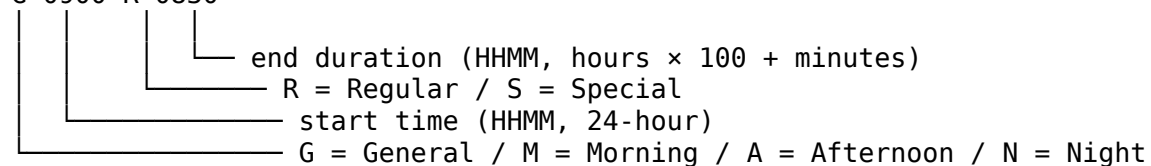
Shift Type naming conventions

This is the single most important convention to internalize. Every Shift Type in Haritha follows a strict 10-character code: `[GMAN]\d{4}[RS]\d{4}` — four-part name where each part is meaningful.

Source of truth: masters/shift_type.csv (25 records currently).

Format breakdown

G 0900 R 0830



Part	Position	Values	Meaning
Prefix	char 1	G / M / A / N	Time-of-day category
Start time	chars 2-5	HHMM (e.g., 0900 = 09:00)	When shift begins
Mid suffix	char 6	R / S	Regular vs Special end
End duration	chars 7-10	HHMM (e.g., 0830 = 8h 30m)	How long the shift lasts

Prefix meanings

Prefix	Time-of-day	Typical use
G	General (daytime, mid-morning start)	Admin, billing, OPD desks
M	Morning (early start)	Doctors, nurses morning shift, ward staff
A	Afternoon (midday start)	Afternoon ward shift, pharmacy afternoon
N	Night (evening/night start, may end past midnight)	ICU night, security, emergency

Mid suffix meanings

Suffix	Meaning
R	Regular shift — standard duration (6h / 8h / 8.5h / 9h)
S	Special shift — extended duration (12h / 16h / 12.5h)

Worked examples (from masters/shift_type.csv)

Name	Prefix	Start	Mid	Duration	End	Use
G0900R0830	General	09:00	Regular	8h 30m	17:30	Standard 9-to-5:30 office shift
G0930R0800	General	09:30	Regular	8h 00m	17:30	Office shift with half-hour late start
M0600R0830	Morning	06:00	Regular	8h 30m	14:30	Early ward / ICU morning
M0800S1200	Morning	08:00	Special	12h 00m	20:00	Long-day shift (e.g., 12h duty)
A1200R0830	Afternoon	12:00	Regular	8h 30m	20:30	Afternoon ward / pharmacy
A1300S1230	Afternoon	13:00	Special	12h 30m	25:30 (next day 01:30)	Long afternoon duty
N2000R1200	Night	20:00	Regular	12h 00m	08:00 next day	ICU night (sets is_past_end_time=1)
N2200R0800	Night	22:00	Regular	8h 00m	06:00 next day	Security late-night
N1700S1600	Night	17:00	Special	16h 00m	09:00 next day	Extended on-call

Why this matters: when the Roster SPA renders the monthly grid, the **color of each cell** is determined by the prefix, and the **start time label** is parsed from the name. So a shift called

G0900R0830 shows up as a blue cell labeled “09:00–17:30” — without anyone hardcoding that mapping.

Color coding (Roster SPA + Tailwind palette)

The HRMS Roster SPA uses **lowercase Tailwind color names** (per LEARNINGS #157 reversal from CapitalCase). The Shift Type’s color field must match a Tailwind palette key:

Prefix	Color (Tailwind)	Hex	Hex code
G (General)	blue	blue-500	#3b82f6
M (Morning)	green	green-500	#22c55e
A (Afternoon)	orange	orange-500	#f97316
N (Night)	violet	violet-500	#8b5cf6

Two Property Setters in `haritha_hospital` fixtures ensure the color dropdown shows the right options: - `Shift Type-color-options` — restricts the Select field to lowercase Tailwind palette names - `Shift Type-color-default` — defaults new Shift Types to blue

If you add a new Shift Type via UI, **always pick from the lowercase dropdown**. Don’t free-type “Blue” — the SPA will crash on `colors['Blue']` lookup because the Tailwind object uses `colors.blue` not `colors.Blue` (LEARNINGS — Roster crash root cause).

Setup workflow (one-time, pre-go-live)

Follow this order. Each step depends on the previous one. Skipping order causes Link field validation errors.

1. Create Company

Already done for Haritha (record name: Haritha Hospitals). Skip if present.

DocType: Company

Name: Haritha Hospitals

Country: India

Default Currency: INR

Fiscal Year: April–March

Custom fields populated by fixtures: - `default_payroll_payable_account` → links to “Salaries Payable” Account - `default_employee_advance_account` → links to “Employee Advances” Account - `default_expense_claim_payable_account` → links to “Expense Claims Payable” Account - `basic_component` → Salary Component “Basic” - `hra_component` → Salary Component “House Rent Allowance” - `arrear_component` → Salary Component “Arrear”

2. Create Departments

Path: HR > Department > New

Required fields:

- Department Name (e.g., "Nursing - ICU", "Housekeeping", "Pharmacy")
- Company: Haritha Hospitals
- Payroll Cost Center: <Cost Center created earlier>

Haritha has **36 departments** (per TRACKER.md and real hospital roster). Examples: - Nursing - ICU - Nursing - Ward A / Ward B - Doctors - General Medicine - Doctors - Emergency - Pharmacy - Laboratory - Radiology - Housekeeping - Security - Administration - Billing - HR - Accounts

3. Create Designations

Path: HR > Designation > New

Required fields:

- Designation Name (e.g., "Staff Nurse", "ICU Specialist", "Pharmacist")

Haritha has **51 designations**. Examples: Staff Nurse, Senior Nurse, ICU Specialist, Consultant, Resident Doctor, Medical Officer, Pharmacist, Lab Technician, Housekeeping Supervisor, Security Guard, Receptionist, HR Executive, Accounts Officer.

4. Create Shift Locations

Path: HR > Shift Location > New

Required fields:

- Location Name (e.g., "Hyderabad Main Hospital")
- Latitude: 17.3850
- Longitude: 78.4867
- Check-in Radius (meters): 200

Shift Location is used by the **HRMS mobile check-in** to verify the employee is physically at the hospital when they punch in. Latitude/longitude is the hospital's geo-coordinates; radius is how far from the pin they can be and still check in (200m allows for parking lot + adjacent wings).

Autoname is field:location_name, so the name == Location Name. Single record for Haritha.

5. Create Shift Types

Path: HR > Shift Type > New

Required fields (CRITICAL – see Lesson #42):

- Name: <10-char code, e.g., G0900R0830>
- Start Time: <HH:MM:SS, e.g., 09:00:00>
- End Time: <HH:MM:SS, e.g., 17:30:00>
- Color: <lowercase Tailwind, e.g., blue>
- Company: Haritha Hospitals
- Holiday List: Haritha Hospitals Holiday List
- last_sync_of_checkin: <FUTURE date, e.g., 2027-01-01 00:00:00> ← REQUIRED (Lesson #42)
- determine_check_in_and_check_out: "Alternating entries as IN and OUT"
- working_hours_calculation_based_on: "First Check-in and Last Check-out"
- begin_check_in_before_shift_start_time: 60 (60 min early grace)
- allow_check_out_after_shift_end_time: 60 (60 min late grace)
- working_hours_threshold_for_half_day: 5.0 (worked <5h = half day)
- mark_auto_attendance: 1 (enabled)

Create all 25 from masters/shift_type.csv via the migration script (see Bulk operations below).

Why last_sync_of_checkin matters: process_auto_attendance() (HRMS background job) silently skips shifts where this is NULL. Symptom is “scheduler runs every minute but marks ZERO attendance”. Always populate it on creation (Lesson #42).

6. Create Holiday List

Path: HR > Holiday List > New

Name: Haritha Hospitals Holiday List

Company: Haritha Hospitals

From Date: 2026-01-01

To Date: 2026-12-31

Holidays (child table):

- 2026-01-26 Republic Day
- 2026-03-25 Holi
- 2026-04-14 Dr. Ambedkar Jayanti
- 2026-08-15 Independence Day
- 2026-10-02 Gandhi Jayanti
- 2026-11-08 Diwali
- Plus weekly offs as configured in HR Settings

This list is then linked to each Shift Type and each Employee — when an employee is scheduled on a date that's a Holiday, Attendance is marked "Holiday" instead of "Absent".

7. Onboard Employees

Path: HR > Employee > New

Required:

- First Name, Last Name, Gender, Date of Birth
- Date of Joining
- Department, Designation, Employment Type
- Company: Haritha Hospitals
- Holiday List: Haritha Hospitals Holiday List
- Default Shift: <one of the 25 Shift Types>
- Attendance Device ID: <EMP-NNNN format, used by biometric>
- Leave Approver: <User>
- Shift Request Approver: <User>

Create all 210 from masters/employee.csv via the migration script. Each Employee links to:
- One Department - One Designation - One Default Shift (used by Auto Attendance cron when no Shift Assignment exists) - One Holiday List - Up to three approver Users (leave / expense / shift)

Daily operations (recurring, every day)

1. Auto Attendance (background)

HRMS scheduler fires `process_auto_attendance_for_all_shifts` **every minute** (configured in `hooks.py` `scheduler_events`). The job:

1. Reads `Shift Type.last_sync_of_checkin` — picks up from where it left off
2. For each Employee with a Shift Assignment, scans Employee Checkin rows between that timestamp and now
3. Pairs IN/OUT entries (alternating) → computes working hours
4. Creates / updates Attendance row:
 - First check-in $\geq$ `start_time` → Present
 - Worked hours $<$ `working_hours_threshold_for_half_day` (5.0) → Half Day
 - No check-ins → Absent (if past `process_attendance_after` date)
 - Date is on Holiday List → Holiday

- Date is in approved Leave Application → On Leave
5. Updates last_sync_of_checkin to now

No manual action needed unless an Employee missed check-in (next step).

2. Manual Attendance (corrections)

When the biometric fails, the employee forgets, or the network drops, the manager marks attendance manually:

Path: HR > Attendance > New

Fields:

- Employee: HR-EMP-00123
- Attendance Date: 2026-08-29
- Status: Present
- Shift: M0600R0830 (their assigned shift)
- Leave Type: <blank unless on leave>
- Company: Haritha Hospitals

Or **bulk mark** for a whole department via the Attendance tool (HR > Attendance > Mark Attendance).

3. Leave applications (employee-side)

Path: HR > Leave Application > New

Employee submits:

- Leave Type: Casual / Sick / Earned
- From Date, To Date
- Reason (optional)
- Attachment (medical certificate for Sick Leave > 2 days)

Approver gets email + desk notification → clicks Approve / Reject
Approved leave automatically reflects in Attendance (status = "On Leave")

The Leave Approver is the User set in Employee.leave_approver (custom field). Default fall-back is the Reporting Manager.

4. Roster view

Path: /app/hr/roster (HRMS Desk page) OR <https://pberpprod.duckdns.org/hr/roster>

The Roster page is a Vue SPA (apps/hrms/roster/) that renders a month grid:

	Aug 2026						
	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Nurse A	[M]	[M]	[A]	[A]	[N]	[N]	[–]
Nurse B	[G]	[G]	[G]	[–]	[G]	[G]	[G]
Nurse C	[–]	[M]	[M]	[M]	[M]	[M]	[M]
...							

Each cell is colored by the Shift Type's color field: - [M] Morning shift → green cell - [A] Afternoon shift → orange cell - [N] Night shift → violet cell - [G] General → blue cell - [–] No shift assigned (grey)

Filter by **Department** (top dropdown) to see only one unit. Filter by **Employee** (search box) to focus on one person.

Diagram

Figure 1: Diagram

Critical lesson (Lesson #69): the Roster shows ONLY rows from tabShift Assignment. If Employee.default_shift is set but no Shift Assignment exists for that date, the Roster shows BLANK — it does NOT fall back to default_shift. Default_shift is only used by the Auto Attendance cron as a fallback for the Attendance computation, NOT the Roster display.

5. Shift swaps (Shift Request)

When two nurses want to swap shifts:

Employee A creates Shift Request:

- Shift Type: M0600R0830
- From Date: 2026-09-05
- To Date: 2026-09-05
- Reason: Personal

Approver (Employee.shift_request_approver) approves

HR runs Shift Assignment Tool to reassign A's M0600 to B for that date

Approval flow is a Custom Field link: Employee.shift_request_approver → User.

Payroll flow

Attendance feeds payroll. Here's the chain:

Step-by-step:

1. **End of month** — HR reviews Attendance + Leave. Fix any anomalies (manually mark attendance for missed check-ins).
 2. **Create Payroll Entry** — HR > Payroll Entry > New. Select Company, Branch, Department, Payable Account (from Company.default_payroll_payable_account), Posting Date.
 3. **Click "Create Salary Slips"** — the system iterates each Employee in the selected department/branch:
 - Reads their Salary Structure
 - Counts days Present / Absent / On Leave / Holiday
 - Pro-rates Basic + HRA based on working days
 - Subtracts PF / ESI / TDS / Loan EMI
 - Generates Salary Slip
 4. **Submit Payroll Entry** — creates Journal Entry that posts:
 - Dr Salary Expense (per Cost Center from Department.payroll_cost_center)
 - Cr Salaries Payable (from Company.default_payroll_payable_account)
 5. **Bank transfer** — separately, Accounts team exports net payable from Salary Slips and processes bank file.
-

Common operations (how-to)

How to add a new Employee

Path: HR > Employee > New

Fill in:

- First Name, Last Name, Gender, Date of Birth
- Date of Joining: today's date
- Status: Active
- Department: <pick from list>
- Designation: <pick from list>
- Employment Type: Full-time
- Company: Haritha Hospitals
- Holiday List: Haritha Hospitals Holiday List
- Default Shift: <their shift, e.g., M0600R0830>
- Attendance Device ID: EMP-NNNN (from biometric vendor)
- Leave Approver: <User who approves leave>
- Shift Request Approver: <User who approves shift swaps>

Save → Submit

The Employee is now visible on the Roster (after a Shift Assignment is created) and eligible for attendance processing.

How to change an Employee's default shift

Path: HR > Employee > <HR-EMP-NNNNN> > Edit
Change "Default Shift" field
Save

Effect: future Shift Assignments will use the new default. Existing Shift Assignments unchanged

To change the actual scheduled shift for a date range, use Shift Assignment (below).

How to bulk-assign shifts

Use the `migrate_master_data.py` script (per Lesson #157 pattern) to load all 8,118 Shift Assignments at once. The script:

1. Reads JSON dump from `/tmp/prod_Shift_Assignment.json`
2. For each row:
 - Checks `frappe.db.exists("Shift Assignment", name)`
 - If exists → `get_doc().update().save()`
 - If missing → `get_doc({"doctype": "Shift Assignment", **payload}).insert()`
3. Remaps the employee field by looking up via `attendance_device_id` (because prod employee IDs differ from dev IDs)

This pattern is **idempotent** — safe to re-run. See `scripts/migrate_master_data.py` for full implementation.

To bulk-assign a SINGLE shift type to MANY employees (e.g., “next week everyone is on Morning shift”), use the **Shift Assignment Tool**:

Path: HR > Shift Assignment Tool
Select: Department, Shift Type, From Date, To Date
Click "Assign"

This creates one Shift Assignment per Employee in the department for each day in the range.

How to handle missed attendance

Symptom: Employee was on shift but no Attendance row exists for today.

Cause 1: Biometric failed (no Employee Checkin)

- Manually mark Attendance via HR > Attendance > New
- Or have employee check-in via HRMS mobile / web

Cause 2: Employee was on approved Leave

- Attendance auto-marks as "On Leave" if Leave Application exists
- If missing, manually create Leave Application (retroactive) and submit

Cause 3: Holiday that day

- Attendance auto-marks as "Holiday"
- Check Shift Type.holiday_list is set

Cause 4: Shift Type.last_sync_of_checkin is in the future

- process_auto_attendance() is skipping this shift (Lesson #42)
- Update last_sync_of_checkin to a past date

How to approve leave

Path: HR > Leave Application > <LA-NNNNN>

Review:

- Employee name, dates, leave type
- Leave balance (shown on form)
- Reason + attachment

Click "Approve" or "Reject" (with reason in dialog)

- If approved, Attendance records for those dates become "On Leave"
- If rejected, employee gets email notification

The approver is determined by Employee.leave_approver. If empty, falls back to Leave Approver from HR Settings.

Edge cases

Holiday during scheduled shift

Employee is scheduled on a Shift Assignment for a date that's on the Holiday List.

Behavior: Attendance = "Holiday" (no Present/Absent). Salary for that day = full Basic (paid holiday).

Why it works: process_auto_attendance() checks Shift Type.holiday_list and Employee's holiday_list first; if the date matches a Holiday, status is forced to "Holiday" regardless of check-ins.

Leave overlap with shift

Employee is on approved Leave AND has a Shift Assignment for the same date.

Behavior: Attendance = “On Leave” (Leave wins over Shift). Salary for that day = full Basic (paid leave).

Why: Leave Application creates a higher-priority record that the Auto Attendance cron checks after Holiday but before Shift.

Shift swap requests

Two employees want to swap a specific date. They use **Shift Request** (not direct Shift Assignment edit).

Step 1: Employee A creates Shift Request for the date they want OFF

Step 2: Approver approves

Step 3: HR manually creates Shift Assignments:

- A's shift on <date>: changed to a different shift type (or removed)
- B's shift on <date>: changed to A's original shift type

Direct editing of submitted Shift Assignments is blocked by Frappe; you must cancel and recreate.

Employee resignation

Step 1: HR > Employee > <HR-EMP-NNNNN> > Edit

Set:

- Status: Left (was Active)
- Relieving Date: <last working day>
- Reason for Leaving: <text>

Save → Submit

Step 2: Cancel all future Shift Assignments for this employee

(HR > Shift Assignment > filter by employee + end_date >= today > Cancel each)

Step 3: Process final settlement:

- Generate final Salary Slip
- Settle Leave Balance (encashment or lapse per policy)
- Recover any outstanding loans
- Issue Experience Letter via Print Format

Step 4: Optionally disable User account (User > <user> > Enabled = 0)

The Employee record is NOT deleted — it's kept for historical payroll, attendance, and audit queries. Only Status changes.

Custom touchpoints (what haritha_hospital fixtures add)

The 274 customizations (per 01.1) that affect shift management:

Custom Fields on Employee

Field	Effect
default_shift	Sets the Employee's fallback shift for Auto Attendance

Field	Effect
leave_approver	Routes Leave Applications to this User
shift_request_approver	Routes Shift Requests to this User
expense_approver	Routes Expense Claims (separate flow)
payroll_cost_center	Per-employee cost center override (used by Payroll Entry)
branch	Org-level branch (defaults from Department)
holiday_list	Per-employee Holiday List (overrides Shift Type default)
attendance_device_id	Biometric ID (EMP-NNNN format), used as FK for checkin matching

Custom Fields on Company

Field	Effect
default_payroll_payable_account	GL account for net Salary payable
default_employee_advance_account	GL account for Employee Advance postings
default_expense_claim_payable_account	GL account for pending Expense Claim reimbursements
basic_component / hra_component / arrear_component	Salary Component references used by Payroll formulas

Custom Fields on Department

Field	Effect
payroll_cost_center	Cost Center for this department's payroll postings
leave_block_list	(Optional) Block leave during certain periods

Custom Fields on Shift Type

Field	Effect
is_oncall	Tag for on-call shifts (used in custom reports)
is_emergency	Tag for emergency shifts

(Note: is_oncall / is_emergency were not in the 78 CF set — check 01.1 for current state. The standard Shift Type color field is what drives Roster SPA rendering.)

Property Setters

Property	DocField	Effect
options	Shift Type-color	Restricts dropdown to lowercase Tailwind palette names (prevents the Roster crash from Lesson #157)
default	Shift Type-color	Defaults new Shift Types to blue

Notification

A Training Scheduled notification fires when a Training Event is created (HRMS standard, re-tagged as Haritha).

Troubleshooting

Symptom	Likely cause	Fix
Roster shows blank for active employees	No Shift Assignments exist for the date range — default_shift does NOT fall back to Roster (Lesson #69)	Create Shift Assignments via Shift Assignment Tool or migrate script
Roster SPA crashes with “Cannot read properties of undefined”	Shift Type color is missing or uppercase (e.g., “Blue” vs “blue”)	Set Shift Type.color to lowercase Tailwind name: blue / green / orange / violet
Auto Attendance marks zero rows despite check-ins	Shift Type last_sync_of_checkin is NULL (Lesson #42)	Set to a future date (e.g., 2027-01-01 00:00:00)
Attendance marked Absent on a holiday	Holiday List not linked to Shift Type or Employee	Set Shift Type.holiday_list OR Employee.holiday_list to “Haritha Hospitals Holiday List”
Leave Approver dropdown empty	Employee.leave_approver not set, no Reporting Manager	Edit Employee > set Leave Approver field
Shift Request approval fails “No Department Approver”	HRMS validate_approver requires Department Approver (Lesson #157 #9)	HR > Department > > add Approver row for the relevant User
Roster filter by Department shows nothing	Employees’ Department field is empty	Edit each Employee > set Department
Payroll Entry creates slips with zero earnings	Salary Structure not assigned to Employee	HR > Salary Structure Assignment > create row linking Employee to Structure
Salary Slip net pay wrong (PF/ESI mismatch)	Statutory components not in Salary Structure	Add PF (Employee + Employer) and ESI (Employee + Employer) Salary Components to Structure

Symptom	Likely cause	Fix
Biometric check-in not creating Attendance	Shift Type mark_auto_attendance = 0	Enable mark_auto_attendance on Shift Type
Background job errors in scheduler log	Custom app Python error in hooks.py scheduler_events	bench --site pberpprod logs → search for traceback → fix in custom app

Diagnostic commands

SSH into VPS as vijay

```
ssh -i /root/.openclaw/ssh_key vijay@144.217.163.228
```

Check scheduler log for Auto Attendance errors

```
docker exec erp-prod-backend-1 tail -100 /home/frappe/frappe-bench/logs/scheduler.log
```

Check count of Attendance records today

```
docker exec erp-prod-db-1 mysql -u root -p<pwd> _b80f05e76a0dcaad \
-e "SELECT COUNT(*) FROM tabAttendance WHERE attendance_date = CURDATE()"
```

Find Shift Types with missing color

```
docker exec erp-prod-db-1 mysql -u root -p<pwd> _b80f05e76a0dcaad \
-e "SELECT name, color FROM tabShift Type WHERE color IS NULL OR color = ''"
```

Check last_sync_of_checkin on a Shift Type

```
docker exec erp-prod-db-1 mysql -u root -p<pwd> _b80f05e76a0dcaad \
-e "SELECT name, last_sync_of_checkin FROM tabShift Type WHERE name = 'G0900R0830'"
```

Quick reference: what to do when

Scenario	First action	Where to look
New employee joins	Create Employee record	HR > Employee > New
Employee quits	Set Status = Left + Relieving Date + cancel future SAs	HR > Employee > Edit
New shift needed	Create Shift Type with 10-char code + lowercase color + last_sync_of_checkin	HR > Shift Type > New
Holiday announced	Add row to Holiday List	HR > Holiday List > Edit
Leave request received	Approve / Reject in Leave Application	HR > Leave Application
Shift swap requested	Approve Shift Request + manually swap SAs	HR > Shift Request + HR > Shift Assignment
Biometric down	Manual attendance for affected employees	HR > Attendance > New
Monthly payroll	Create Payroll Entry → Create Salary Slips → Submit	HR > Payroll Entry
Roster empty	Create Shift Assignments via Tool	HR > Shift Assignment Tool

Scenario	First action	Where to look
Audit “who was on duty X date”	Filter Roster by date	/hr/roster

Related docs

Doc	Purpose
01.1 Schema Reference	Full list of 274 customizations
01.2 Schema Diagram	ERD of shift + HR entities
00.1 What is ERP/Frappe/HRMS	Background on the stack
00.2 Architecture Overview	Container / network / backup
02.2 Process Diagram	Mermaid flowcharts of the same flows

End of 02.1. Next: 02.2 — Process Diagram.